Find the series' radius of convergence.

1) $\sum_{n=0}^{\infty} \frac{(x-2)^n}{2n+1}$

2) $\sum_{n=1}^{\infty} \frac{(x-5)^n}{\ln(n+2)}$

Find the interval of convergence of the series.

3) $\sum_{n=0}^{\infty} \frac{(x-3)^n}{3n+7}$

Find the sum of the series as a function of x.

4) $\sum_{n=1}^{\infty} (x+7)^n$

Provide an appropriate response.

5) If the series $\sum_{n=0}^{\infty} (-1)^n (x-4)^n$ is integrated term by term, for what value(s) of x does the new series converge?

6) Find the sum of the series $\sum_{n=1}^{\infty} \frac{n}{9^{n-1}}$ by expressing $\frac{1}{1-x}$ as a geometric series, differentiating both sides of the resulting equation with respect to x, and replacing x by $\frac{1}{9}$.

Find the Taylor polynomial of order 3 generated by f at a.

7) $f(x) = \frac{1}{7-x}$, $a = 1$

Find the Taylor series generated by f at x = a.

8) $f(x) = \frac{1}{9-x}$, $a = 6$

9) $f(x) = e^x$, $a = 3$

Find the quadratic approximation of f at x = 0.

10) $f(x) = e^{\sin 2x}$

Use substitution to find the Taylor series at x = 0 of the given function.

11) e^{-2x}

Use power series operations to find the Taylor series at x = 0 for the given function.

12) $f(x) = x^3 \sin x$

Find the first four nonzero terms in the Maclaurin series for the function.

13) $f(x) = e^{4x}$

Find the first four terms of the binomial series for the given function.

14) $(1 + 10x)^{1/2}$

Answer Key
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1) 1
2) 1
3) $2 \le x < 4$
4) $-\dfrac{x+7}{x+6}$
5) $3 < x \le 5$
6) $\dfrac{81}{64}$
7) $P_3(x) = \dfrac{1}{6} - \dfrac{x-1}{36} + \dfrac{(x-1)^2}{216} - \dfrac{(x-1)^3}{1296}$
8) $\displaystyle\sum_{n=0}^{\infty} \frac{(x-6)^n}{3^{n+1}}$
9) $\displaystyle\sum_{n=0}^{\infty} \frac{e^3 (x-3)^n}{n!}$
10) $Q(x) = 1 + 2x + 2x^2$
11) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n (2)^n x^n}{n!}$
12) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+4}}{(2n+1)!}$
13) $1 + 4x + \dfrac{16x^2}{2!} + \dfrac{64x^3}{3!} + \ldots$
14) $1 + 5x - \dfrac{25}{2}x^2 + \dfrac{125}{2}x^3$